

Decision Tree vs Random Forest: Overfitting, Pruning and Performance Comparison

When we train a model, two types of errors can occur:

Underfitting

Overfitting

These are explained using:

Bias

Variance

Bias = Error due to wrong assumptions.

High Bias model:

Too simple

Misses important patterns

Underfits

Variance = Sensitivity to training data.

High Variance model:

Changes drastically with small data changes

Unstable

Overfits

Example: Train on slightly different dataset → completely different tree.

Decision Trees → High Variance

You cannot minimize both at the same time.

When:

Bias ↓ → Variance ↑

Variance ↓ → Bias ↑

Total Error:

Error = Bias² + Variance + Irreducible Error

Irreducible error = noise in data (cannot remove)

Overfitting Overfitting happens when model learns:

Real patterns

Noise

Random fluctuations

What happens?

Very low training error

High testing error

Why?

Low Bias

High Variance

Underfitting happens when the model is too simple to learn the pattern in data.

What happens?

High training error

High testing error

Model cannot capture complexity

Why?

High Bias

Low Variance

Memorization = Overfitting

Oversimplification = Underfitting

Decision Tree → Can memorize (overfit)

Random Forest → Reduces memorization using ensemble

A Decision Tree:

Splits data based on features

Creates branches until stopping condition

Works like asking yes/no questions

Example:

Is Age > 30?

Is Income > 50K?

etc.

Problem: Overfitting

If we let tree grow fully:

It creates very deep branches

It memorizes training data

Training accuracy → 100%

Test accuracy → drops

High Variance Model

How Decision Tree Solves Overfitting

1. Pre-Pruning (Early stopping)

Limit max_depth

Min samples split

Min samples leaf

2. Post-Pruning

Grow full tree → then cut weak branches

But still...

Small changes in data → completely different tree

Unstable model

Still high variance

Random Forest

Random Forest = Many Decision Trees + Voting

How it works:

Take random samples of data (Bootstrap)

For each split → consider random subset of features

Build many trees

Final output = Majority Vote

Why Random Forest Reduces Overfitting

Because:

One tree overfits differently

Another tree overfits differently

Averaging cancels errors

This reduces:

Variance

Overfitting

Instability

More stable + Better generalization

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
```

```
data = load_breast_cancer()
X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)

train_pred = dt.predict(X_train)
test_pred = dt.predict(X_test)

print("Decision Tree (No Pruning)")
print("Train Accuracy:", accuracy_score(y_train, train_pred))
print("Test Accuracy:", accuracy_score(y_test, test_pred))
```

```
Decision Tree (No Pruning)
Train Accuracy: 1.0
Test Accuracy: 0.9415204678362573
```

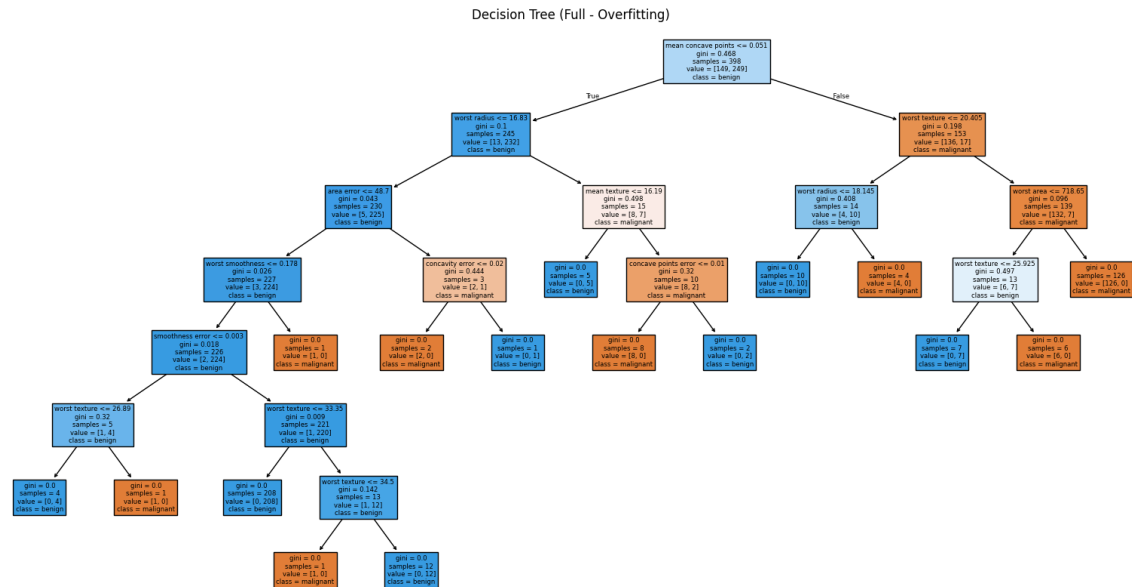
Training Accuracy >> Testing Accuracy

Model is overfitting.

Overfitting = Model performs extremely well on training data But fails to generalize to unseen data.

Generalization is the key

```
plt.figure(figsize=(20,10))
plot_tree(dt,
          feature_names=data.feature_names,
          class_names=data.target_names,
          filled=True)
plt.title("Decision Tree (Full - Overfitting)")
plt.show()
```



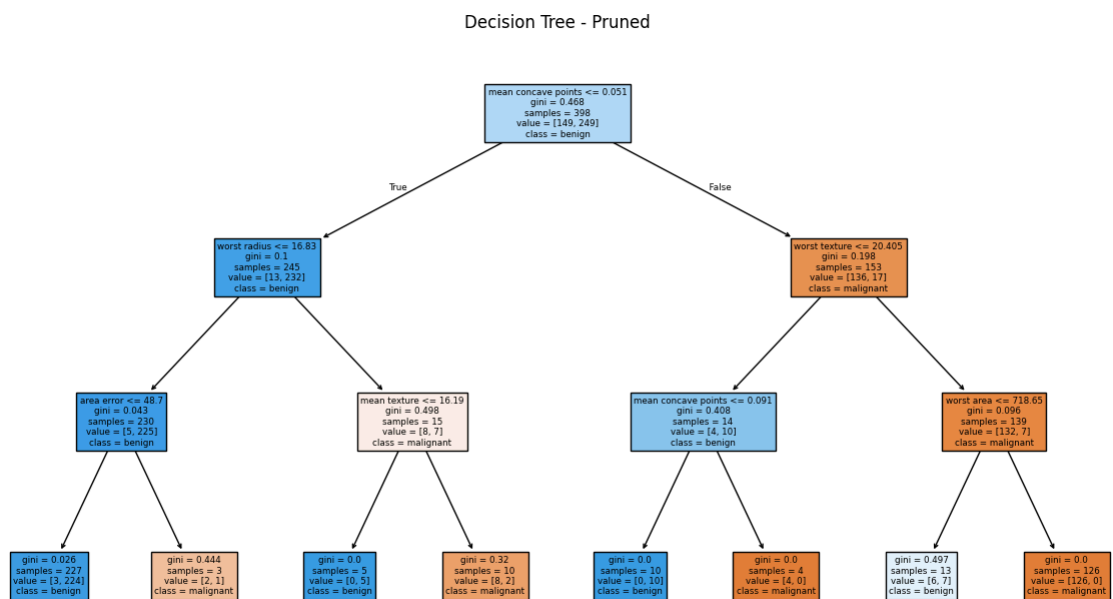
```
dt_pruned = DecisionTreeClassifier(max_depth=3, random_state=42)
dt_pruned.fit(X_train, y_train)

train_pred_p = dt_pruned.predict(X_train)
test_pred_p = dt_pruned.predict(X_test)
```

```
print("Decision Tree (Pruned)")
print("Train Accuracy:", accuracy_score(y_train, train_pred_p))
print("Test Accuracy:", accuracy_score(y_test, test_pred_p))
```

Decision Tree (Pruned)
 Train Accuracy: 0.9698492462311558
 Test Accuracy: 0.9649122807017544

```
# Plot Pruned Tree
plt.figure(figsize=(15,8))
plot_tree(dt_pruned,
          feature_names=data.feature_names,
          class_names=data.target_names,
          filled=True)
plt.title("Decision Tree - Pruned")
plt.show()
```



Ensemble Learning is a technique where:

Multiple models are combined to make a better final prediction.

Instead of relying on one model, we combine many models.

Goal:

Improve accuracy

Reduce overfitting

Improve stability

Bagging (Bootstrap Aggregating)

Example: Random Forest

How it works:

Take random samples of data (with replacement)

Train multiple models

Final output = Average / Majority Vote

Purpose:

Reduce variance

Control overfitting

Used when:

Model has high variance (like Decision Tree)

```
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)

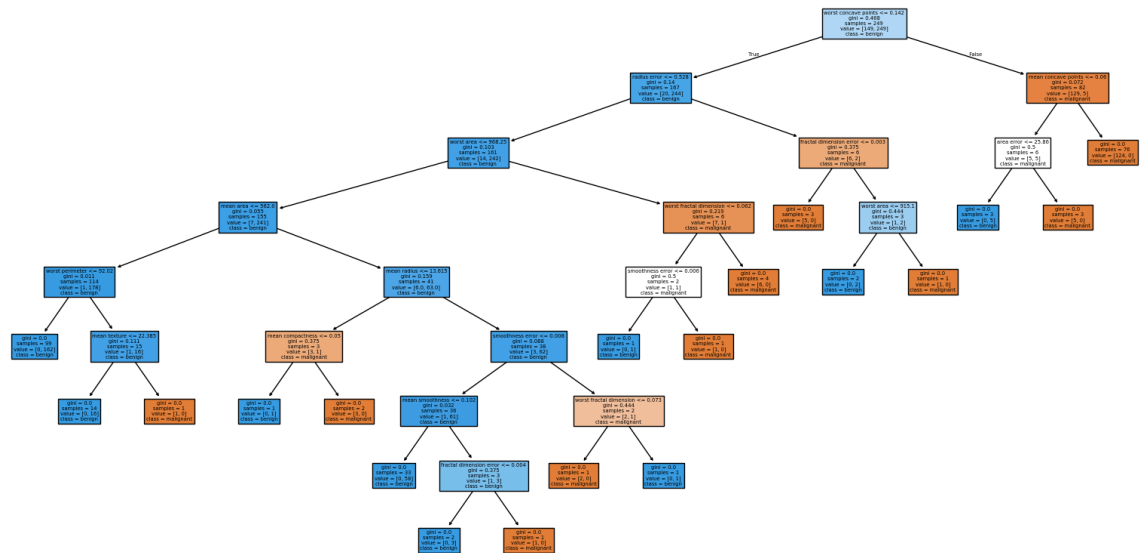
train_pred_rf = rf.predict(X_train)
test_pred_rf = rf.predict(X_test)

print("Random Forest")
print("Train Accuracy:", accuracy_score(y_train, train_pred_rf))
print("Test Accuracy:", accuracy_score(y_test, test_pred_rf))
```

```
Random Forest
Train Accuracy: 1.0
Test Accuracy: 0.9707602339181286
```

```
one_tree = rf.estimators_[0]
plt.figure(figsize=(20,10))
plot_tree(one_tree,
          feature_names=data.feature_names,
          class_names=data.target_names,
          filled=True)
plt.title("One Tree from Random Forest")
plt.show()
```

One Tree from Random Forest



Summary:

Underfitting = High Bias

Overfitting = High Variance

Good Model = Balance of both

Random Forest reduces variance while Pruning controls overfitting

Decision Tree alone suffers from high variance.

Pruning reduces variance but increases bias.

Random Forest smartly reduces variance without increasing bias too much.

That is the power of ensemble learning.

